

Python Practice Handout

Randomness and the Fortune Cookie project

1. Importing a module

A module is a chunk of code that someone else has already written. Python has many of them, but you only load the ones you need. To use randomness, import the random module at the top of your program.

```
import random
```

2. A random whole number

Write the module name, a dot, then randint. Give it a smallest and a largest number. It hands back a random whole number between the two.

```
import random  
random.randint(1, 10)
```

Run this and nothing appears. As always, you have to print it to see it.

```
import random  
print(random.randint(1, 10))
```

Example output (yours will differ):

```
8
```

Run it several times. You get a different number each time.

3. A random decimal number

There is also random.random, with nothing inside the brackets. It gives back a random float between 0 and 1.

```
import random  
print(random.random())
```

Example output (yours will differ):

```
0.6654312988017358
```

4. Looking things up

You are not expected to memorise names like randint. Search the web for python random and you will find pages listing everything the module can do. Looking things up is a normal part of programming.

5. Try it yourself

Make a magic 8 ball that prints Yes, No or Maybe each time you run it. Pick a random number from 1 to 3 and use an if statement for each one. Try this on your own before looking below.

```
import random

answer = random.randint(1, 3)

if answer == 1:
    print("Yes")
if answer == 2:
    print("No")
if answer == 3:
    print("Maybe")
```

Example output (yours will differ):

```
Maybe
```

Remember the two equal signs when comparing, and remember to tab the print line across.

6. The project: start with the end in mind

Now build a fortune cookie. First decide what you want the user to see. The user just means the person running your program.

```
print("You will have a great day! Your lucky number is 44")
```

Output:

```
You will have a great day! Your lucky number is 44
```

This works, but it says the same thing every time. The fortune and the number both need to change.

7. A random lucky number

Import random, store a number from 1 to 100 in a variable, then put that variable into the sentence. Turn the string into an f-string so the curly brackets work.

```
import random

lucky_number = random.randint(1, 100)

print(f"You will have a great day! Your lucky number is {lucky_number}")
```

Example output (yours will differ):

```
You will have a great day! Your lucky number is 51
```

8. A random fortune

Do the same trick for the fortune itself. Pick a number from 1 to 3, start with an empty string, then use if statements to fill that string with one of three fortunes.

```
import random

fortune_number = random.randint(1, 3)
fortune_text = ""

if fortune_number == 1:
    fortune_text = "You will have a great day!"
if fortune_number == 2:
    fortune_text = "Today will be tough... but worth it."
if fortune_number == 3:
    fortune_text = "You will get married this year!"

print(fortune_text)
```

Example output (yours will differ):

```
Today will be tough... but worth it.
```

9. The finished program

Put both halves together and place each variable in the f-string. This one program uses a module, a random number, a string, an if statement and an f-string all at once.

```
import random

lucky_number = random.randint(1, 100)

fortune_number = random.randint(1, 3)
fortune_text = ""

if fortune_number == 1:
    fortune_text = "You will have a great day!"
if fortune_number == 2:
    fortune_text = "Today will be tough... but worth it."
if fortune_number == 3:
    fortune_text = "You will get married this year!"

print(f"{fortune_text} Your lucky number is {lucky_number}")
```

Example output (yours will differ):

```
You will have a great day! Your lucky number is 89
Today will be tough... but worth it. Your lucky number is 81
You will get married this year! Your lucky number is 2
```

Run it a few times and you should see all three fortunes appear.

10. Try it yourself

Add a fourth fortune of your own. You will need to change the largest number in randint and add one more if statement.